DO NOW — RECALL

Bubble sort: trace and implement

Name: _____ Class: _____ Date: _____

1. For CountDigits(582), list the calls and give the returned digit count.

2. How many times does a loop with i = 0; i < 4; i++ run?

3. For A=true and B=false, what is A && B?

4. After swapping neighbouring values 7 and 3 in [7,3,5], what is the array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Calls 582 → 58 → 5; returns 1 → 2 → 3. The digit count is 3.

2. Model answer

4 times: i is 0,1,2,3.

3. Model answer

false. Both must be true for AND.

4. Model answer

[3,7,5].

Vocabulary — know these terms

pass — One traversal of the remaining unsorted section.

adjacent — At neighbouring indices.

swap — Exchange two stored values.

comparison — Test used to determine whether a swap is required.

early exit — Stop when a pass makes no swaps.

NEW LEARNING

Read, annotate and explain

Compare neighbours

Compare positions j and j+1. Swap only if the earlier value is greater.

Finish one pass

The greatest unsorted value moves to the right-hand end of the unsorted section.

Repeat less work

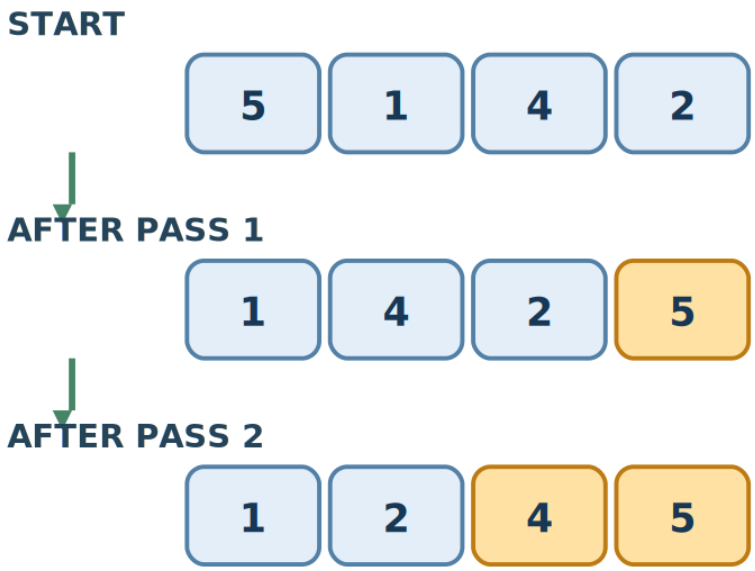
Each completed pass fixes one final position; an entire pass with no swaps proves the array is sorted.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Read the code; predict its effect



Worked algorithm states: START 5,1,4,2; AFTER PASS 1 1,4,2,5; AFTER PASS 2 1,2,4,5

I DO — WORKED EXAMPLE

Follow the worked method

Trace a complete pass of ascending bubble sort on [6,2,5,1].

Predict first

Compare (6,2), (6,5), then (6,1).

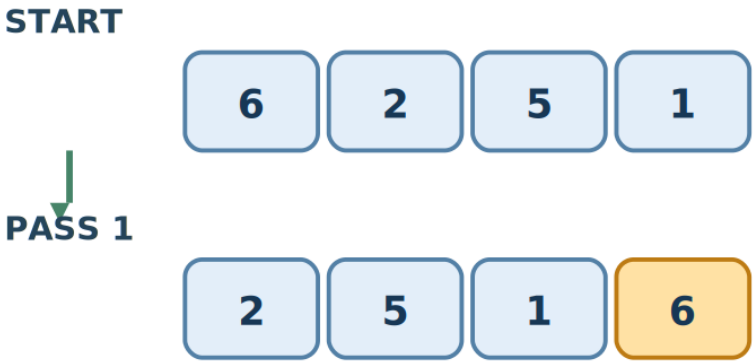
Model solution

After swaps: [2,6,5,1] → [2,5,6,1] → [2,5,1,6]. The last value is fixed.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a complete pass of ascending bubble sort on [6,2,5,1].



Worked algorithm states: START 6,2,5,1; PASS 1 2,5,1,6

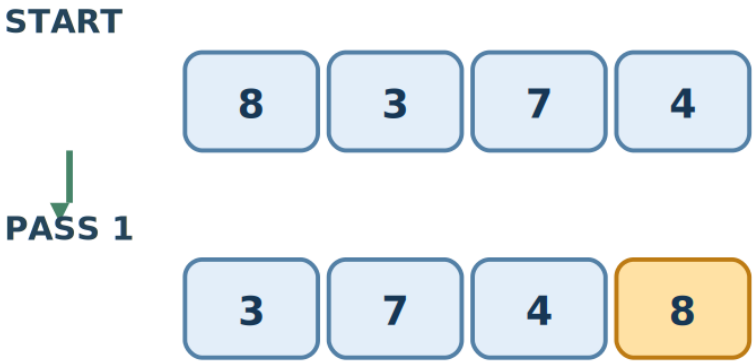
WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Complete one pass on [8,3,7,4].



Worked algorithm states: START 8,3,7,4; PASS 1 3,7,4,8

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Which adjacent pair is compared first?

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Show one left-to-right pass on [7,3,6,2].

Question 2

Show two complete passes on [7,3,6,2].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Why can the early-exit optimisation finish immediately on [1,2,3,4]?

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Bubble sort always needs to make at least one swap.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model solution

8 with 3, then 8 with 7, then 8 with 4. The result is [3,7,4,8].

IMPROVE: Explain one correction to your own reasoning.